Mahmoud Bekri PhD, CIB

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BACKGROUND

Mahmoud Bekri is a Research Economist at the Islamic Development Bank Institute (IsDBI). He holds a PhD in Economics and Finance and a German Diploma in Physics (equivalent to a bachelor's and a master's degree combined), both from the Karlsruhe Institute of Technology (KIT) in Germany, where he subsequently worked as a Postdoctoral Researcher at the Institute for Economic Theory and Statistics.

With a strong foundation in mathematics and physics, his advanced quantitative and analytical skills deepen his expertise in economics and finance. His career combines academic and professional work: he has taught advanced courses in financial engineering, contributed to research in Islamic, ethical and social finance, and advised international banks and asset management firms on advanced financial methodologies. His earlier roles include strategist, project manager and sales manager. He publishes regularly in peer-reviewed journals.

Since joining the IsDB Group in 2015 through the Young Professionals Programme, he has played a key role in numerous projects and led global initiatives to advance the Islamic finance sector across several departments, including the Islamic Financial Services Department, the Financial Product Development Center, IRTI and, most recently, the IsDB Institute. He earned the Certified Islamic Banker (CIB) designation in 2016, the Leadership Certificate in Islamic Banking and Finance in 2018 and the Certificate in Advanced Project Management in 2024.

EDUCATION

2014 — **PhD in Economics and Finance**
Karlsruhe Institute of Technology (KIT), Germany · Institute of Economics (ECON), Chair of Econometrics and Statistics
"Portfolio Optimization and Risk Management under Non-normal Stable and Tempered Stable Distributions in Islamic Finance"
Magna cum laude · Supervisors: Prof. Svetlozar Rachev (Stony Brook University, US), Prof. Wolf-Dieter Heller (KIT), Prof. Frank Fabozzi (Yale University, US)

2011 — **German Diplom in Physics (Diplom-Physiker)**
Karlsruhe Institute of Technology (KIT), Germany
"Studies of Si-MCM-41 and Al-MCM-41 Surface Structure and Surface Charge"
Final grade "very good" (sehr gut) · Supervisor: Prof. Schimmel

Professional certificates

2024 — **Certificate in Advanced Project Management**
Meirc Professional Certificate (MPC), member of PMI · financed by employer

2018 — **Leadership Certificate in Islamic Banking and Finance**
Cambridge Islamic Finance Leadership Programme (IFLP) 2018 · financed by employer

2016 — **Certified Islamic Banker (CIB)**
CIBAFI · financed by employer

PROFESSIONAL EXPERIENCE

01/2026–present — **Secondment, Model Risk**
Risk Management Department, IsDB

- Contribute to building the Bank's model risk management framework.

- Designed a multi-manager equity risk oversight framework that monitors external equity managers against their benchmark (tracking error, risk attribution and performance assessment), documented for formal model validation.
- Built an asset-liability management cash-flow stress model and an Islamic ALM model, with full reconciliation and quality-control checks.
- Supported hedging analysis, including swap hedge accounting, and delivered an internal workshop on hedging Islamic equity portfolios.

2018–present

Research Economist

Islamic Development Bank Institute (IsDBI)

- Lead research, policy advisory and capacity-building initiatives to advance Islamic finance across IsDB member countries.
- Conduct applied economic and quantitative research on sustainable finance, risk-sharing instruments and fintech innovation.
- Develop analytical frameworks and financial models supporting the design of Islamic financial products and sovereign instruments such as Sukuk.
- Collaborate with central banks, ministries of finance and development partners to strengthen Islamic finance ecosystems and regulatory capacity.
- Represent the Institute at international forums and conferences on Islamic finance, the digital economy and sustainability.
- Supervise and contribute to publications, working papers and knowledge products advancing the Institute's research agenda.

Notable achievements

- Designed and delivered capacity-building programmes for central banks and financial regulators in Libya, Morocco, Uganda and Mauritania.
- Contributed to the development and approval of several Islamic finance grants supporting Awqaf, participatory finance and social finance initiatives across member countries.
- Co-authored research on GDP-linked Sukuk, Islamic financial benchmarks and inflation modelling after COVID-19.
- Played a key role in IsDBI's patented "Smart Voucher System" (US Patent 12282916), applying fintech innovation to development finance.
- Published peer-reviewed papers and thought-leadership blogs on Islamic finance, sustainable development and financial innovation.
- Authored the book *Heavy Tails and Coherent Risk Measures in Islamic Finance* on advanced financial modelling techniques.
- Delivered the capacity-building programme "Introduction to Islamic Finance" for the Interuniversity Postgraduate Programme in Economics (PTCI) in Dakar, Senegal.
- Built the database and performed the quantitative analysis of the Consensus Economics survey data, used to assess the prospects of an Islamic financial returns benchmark for Islamic financial institutions.
- Authored the blogs "Inflation Dependence Structure in a post-COVID-19 World" and "Crypto GDP Sukuk, for Better Debt Servicing in Post-COVID-19", and translated three blogs from English into Arabic.
- Drafted and finalised the IsDBI contribution to the IsDB Group Member Country Partnership Strategies (MCPS) for Niger, Guinea and Morocco.
- Co-authored the paper "Taming Inflation: An Islamic-finance Perspective".
- Represented IsDBI in the Libya Task Force, delivering training to Libyan financial institutions including the Central Bank of Libya and the Libya Development Bank.
- Coordinated the Palestine Monetary Authority Digital Currency Project.

- Member of the organising team of the first Economists' Forum, "The Rise of Central Bank Digital Currency: Motives, Approaches, and Techniques".
- Contributed to translating IsDBI works from English into French and Arabic.
- Nominated technical assistance project officer for projects in Uganda, Burkina Faso and the UAE.
- Authored and presented the manuscript "On the Role of Cryptocurrencies in Islamic Finance, Real Economy and Sustainability" in Istanbul, Türkiye.
- Led a task force and performed the quantitative work on "Developing an Islamic Financial Returns Benchmark for Islamic Financial Institutions".
- Contributed to obtaining two patents, "The Blockchain-based Voucher System" and "Blockchain-based Credit Management".
- Authored and translated blogs on contemporary issues in economics, finance, fintech and sustainable development.
- Designed a fintech product to support sovereign Islamic finance initiatives.
- Worked on the Smart Economy Project to patent and implement a fintech product.
- Presenter and panellist at the ADFIMI and Qatar Development Bank joint workshop on "Sharia Compliance Issues of Crypto Assets", Qatar.
- Member of the Center of Excellence on Islamic Finance team, which organised its first workshop in Madinah.
- Submitted the research paper "The Determination of Alpha in CAR" to the Tunisia International Conference on Islamic Finance (TICIEF 2019).
- Conducted an international workshop on the challenges of capital regulation for Islamic financial institutions in the Sultanate of Oman.
- Evaluator of the Queen Alia Airport Project in Amman, Jordan, on a joint mission with the IDB Operations Evaluation Department (OED).
- Pricing analyst and product developer in the team that designed the Sukuk Assets Utility Awqaf.
- Authored "The Determination of Alpha in the Capital Adequacy Ratio for Islamic Financial Institutions".
- Authored a case study of the Islamic Banking Africa Summit as an advocacy platform.
- Co-editor of the Islamic Finance Country Report on Djibouti.

2015–2018

Young Professional

Islamic Development Bank Group, Young Professionals Programme (YPP)

The YPP is the strategic talent pipeline for professional careers in the IDB Group, combining a training programme with hands-on rotations.

Financial Product Development Center, third rotation

- Represented the department in the IDB Group Risk Management Committee.
- Authored the "Survey, Study & Analysis of the Financial Structures Depending on GDP Shares, Commodities Linked-Shares & Similar Structures".
- Authored the first part of the study "Fiduciary and Third-Party Assessment Risk".
- Took part in the design of the Sovereign Finance Company (SFC) and its cash-flow model.
- Took part in the Development Effectiveness Review of the IsDB.
- Analysed the corporate and US Treasury bonds used to price Sukuk.

Financial Product Development Center, second rotation

- Represented the department in the IDB Group Risk Management Committee.
- Attended the 29th International Summer School of the Swiss Association of Actuaries on Quantitative Risk Management, Lausanne.
- Authored a survey of credit default swaps (CDS) and a brief survey of CDS modelling.
- Determined the premiums of performance-sensitive financial papers.
- Drafted a first export index for the exports of OIC member countries.

- Optimised the stock basket of exchangeable Sukuk taken up by Société Générale.
- Built a Sukuk databank and analysed the tail risk of tradable Sukuk.

Islamic Financial Services Department (IFSD), first rotation

- Audited the Saudi Project for Hajj Meat (Adahi) managed by the IDB.
- Presented the Zitouna Tamkeen Project to the investment committees and prepared the due diligence questionnaire and the final project report.
- Contributed to the Technical Assistance (TA) process review, including the team meetings and the drafting of the TA request form.
- Reviewed the financial and legal feasibility study of the Awqaf Properties Investment Fund Project in Cotonou, Benin, took part in the appraisal mission and the meetings with the consultants and the Ministry of Finance, and helped refine the project costing.
- Contributed to the Islamic Financial Services Board Council meeting.
- Prepared the presentation on the IF Gates in Kuwait and Luxembourg.
- Contributed to the committee establishing the corporate identity and marketing of the IFS Department's activities.

11/2013–02/2015 **Postdoctoral Researcher**
Institute for Economic Theory and Statistics (ETS), KIT, Germany

- Developed research on financial engineering, portfolio optimisation and risk management.
- Managed the research projects on Islamic finance.
- Supervised bachelor's and master's theses and co-supervised a PhD thesis.
- Taught courses on financial engineering and advanced financial methods, and seminars on hedging and financial products.

11/2013–09/2014 **Senior Consultant, Islamic finance projects**
FinAnalytica and Bravo Risk Management, New York

- Developed investment portfolio analytics and asset management tools.
- Improved the performance of long-term investment portfolios with tail-risk analytics.
- Optimised investment portfolios, enhancing performance and mitigating risk.
- Applied stable distributions to the components of the German DAX index, and heavy-tailed distributions to Sukuk.
- Developed a risk management software tool for daily trading in Islamic finance.

04/2013–10/2013 **Senior Social Network Analysis (SNA) Analyst and Consultant**
SONEAN GmbH (powered by Funds@Work), Frankfurt, Germany

- Developed SNA-based business strategies, and supported and followed up their implementation.
- Created and continuously extended the SONEAN database.
- Systematically analysed the social capital and networks of decision-makers.
- Identified business risks and opportunities for clients.

12/2012–02/2013 **Sales Manager**
Edmund Optics, Karlsruhe, Germany · leading producer of optics, imaging and photonics technology

- Provided customer- and application-oriented technical advice.
- Performed feasibility analyses and calculated offers.
- Sold customised products and processed delivery requests.

09/2012–11/2012

Intern, Portfolio Pricing Department

Hoist Finance Group, Düsseldorf, Germany

- Gained an overview of the methods and challenges of rating unsecured non-performing loans (NPL), and of the phases of debt and receivables management.
- Priced NPL portfolios and determined recovery and cost curves.
- Created and presented the portfolio assessment template for the investment committee on portfolio purchases.

03/2011–08/2012

Project Manager

TISFO (Tunisian Industrial Services for Oil), Sousse, Tunisia

- Carried out market research and competitive analysis.
- Managed project delivery: planning, implementation and final control.
- Strengthened business relations with existing and new customers.
- Coordinated with departments and stakeholders to meet tight deadlines, and controlled the budget.

SELECTED PUBLICATIONS

Book and book chapter

Bekri, M. (2021). *Heavy Tails and Coherent Risk Measures in Islamic Finance.* Islamic Development Bank Institute, Jeddah. 170 pp. ISBN 978-603-8306-06-2.

Bekri, M., Kabir Hassan, M., Alam, N. (2017). On the dependency structure of Islamic assets. In M. Kabir Hassan (ed.), *Handbook of Empirical Research on Islam and Economic Life*, pp. 602–623.

Doctoral thesis

Bekri, M. (2014). *Portfolio Optimization and Risk Management under Non-normal Stable and Tempered Stable Distributions in Islamic Finance.* Doctoral thesis, KIT, Germany.

Indexed journals

Hajjar, M., Bekri, M. (2026). Quick Commerce and Islamic Ethics: Maqāṣid and Policy Assessment. *International Journal of Islamic Finance and Sustainable Development*, 18(3), 41–59. doi:10.55188/ijifsd.v18i3.1461

Mirzaei, M., Bekri, M. (2017). Energy consumption and CO_2 emissions in Iran, 2025. *Environmental Research*, 154, 345–351.

Mirzaei, M., Bekri, M., Ramakrishnan, S. (2016). Corporate Default Prediction with Industry Effects: Evidence from emerging markets. *International Journal of Economics and Financial Issues*, 6(S3), 161–169.

Ramakrishnan, S., Mirzaei, M., Bekri, M. (2015). Corporate Default Prediction with Adaboost and Bagging Classifiers. *Jurnal Teknologi*, 73(2), 45–50.

Ramakrishnan, S., Mirzaei, M., Bekri, M. (2015). A Multi-Industry Default Prediction Model using Logistic Regression and Decision Tree. *Research Journal of Applied Sciences, Engineering and Technology*, 9(10), 856–861.

Ramakrishnan, S., Mirzaei, M., Bekri, M. (2015). Adaboost Ensemble Classifiers for Corporate Default Prediction. *Research Journal of Applied Sciences, Engineering and Technology*, 9(3), 224–230.

Bekri, M., Kim, Y. S. (2015). Tail risk analysis of the S&P/OIC COMCEC 50 index. *Borsa Istanbul Review*, 15(1), 1–16.

Bekri, M., Kim, Y. S. (2014). Portfolio management with heavy tailed distributions in Islamic Finance. *Journal of Islamic Economics, Banking and Finance*, 10(2), 114–137.

Bekri, M., Kim, Y. S., Rachev, S. T. (2014). Tempered stable models for Islamic finance asset Management. *International Journal of Islamic and Middle Eastern Finance and Management*, 7(1), 37–60.

Conference papers

Bekri, M. (2026). Pricing Catastrophe-Tailed Energy Assets: A Stable-Law Expected-Shortfall Correction to the Cost of Capital for Nuclear Power. Accepted, 4th IAEE MENA Conference, KAPSARC, Riyadh, 7–8 December 2026.

REFEREE FOR

2015	*International Journal of Islamic and Middle Eastern Finance and Management*
2014	*Frontiers in Mathematical Finance*

LANGUAGES AND SOFTWARE

Languages	Arabic (mother tongue) · English, French and German (fluent)
Software	MATLAB and Python (frequent user) · MS Office: Word, Excel, PowerPoint, Access (good)
Interests	Reading, writing, sport and travelling

OTHER ACTIVITIES

- Member of the German Business Circle of the German-Saudi Arabian Liaison Office for Economic Affairs, Jeddah, Saudi Arabia.
- Member of the Tunisian Scientific Society (TWG), Karlsruhe, Germany.
- Member of the academic “Wednesday Talk”, a meeting of economic scholars at King Abdulaziz University, Jeddah, Saudi Arabia.
- Former Chairman of the Jury of the Master of Islamic Financial Management, Effat College of Business, Jeddah, Saudi Arabia.